

Task 1

```
postgres=# create table employees(department varchar(50), salary numeric);
CREATE TABLE
postgres=# INSERT INTO employees (department, salary)
VALUES
('IT', 75000),
('HR', 55000),
('Finance', 68000),
('Marketing', 62000),
('Sales', 58000),
('Operations', 64000);
INSERT 0 6
```

```
postgres=# select department, avg(salary) from employees group by department having avg(salary) > 60000;
 department |          avg
-----+-----
 Marketing  | 62000.000000000000
 Operations | 64000.000000000000
 Finance    | 68000.000000000000
 IT         | 75000.000000000000
(4 rows)
```

```
postgres=# select department, avg(salary) from employees group by department
postgres-# ;
 department |          avg
-----+-----
 Marketing  | 62000.000000000000
 Operations | 64000.000000000000
 Finance    | 68000.000000000000
 Sales      | 58000.000000000000
 IT         | 75000.000000000000
 HR         | 55000.000000000000
(6 rows)
```

Task 2

```
postgres=# create table sales(sales_rep varchar(50), sales_amount numeric);
CREATE TABLE
postgres=# INSERT INTO sales (sales_rep, sales_amount)
postgres=# VALUES
postgres=#      ('Alice', 5000),
postgres=#      ('Bob', 7200),
postgres=#      ('Alice', 6500),
postgres=#      ('Charlie', 4300),
postgres=#      ('Bob', 8100),
postgres=#      ('Charlie', 5500);
INSERT 0 6
```

```
postgres=# select sales_rep, sum(sales_amount) from sales group by sales_rep;
```

sales_rep	sum
Alice	11500
Bob	15300
Charlie	9800

(3 rows)

```
postgres=# select sales_rep, sum(sales_amount) from sales group by sales_rep having sum(sales_amount) between 20000 and 40000;
```

sales_rep	sum
-----------	-----

(0 rows)

```
postgres=# select sales_rep, sum(sales_amount) from sales group by sales_rep having sum(sales_amount) between 5000 and 20000;
```

sales_rep	sum
Alice	11500
Bob	15300
Charlie	9800

(3 rows)

Task 3

```
postgres=# create table employees(department varchar(50), job_title varchar(50));
CREATE TABLE
postgres=# INSERT INTO employees (department, job_title)
postgres=# VALUES
postgres=#      ('IT', 'Software Engineer'),
postgres=#      ('IT', 'Data Analyst'),
postgres=#      ('IT', 'DevOps Engineer'),
postgres=#      ('HR', 'HR Manager'),
postgres=#      ('HR', 'Recruiter'),
postgres=#      ('Finance', 'Accountant'),
postgres=#      ('Finance', 'Financial Analyst'),
postgres=#      ('Marketing', 'Marketing Manager'),
postgres=#      ('Marketing', 'Content Specialist'),
postgres=#      ('Sales', 'Sales Representative'),
postgres=#      ('Sales', 'Sales Manager'),
postgres=#      ('Sales', 'Account Executive');
INSERT 0 12
```

```
postgres=# select department, job_title, count(*) from employees group by department,job_title;
 department |      job_title      | count
-----+-----+-----
IT          | DevOps Engineer     |      1
IT          | Data Analyst        |      1
HR          | Recruiter           |      1
Finance     | Accountant          |      1
Sales       | Sales Manager       |      1
Sales       | Sales Representative |      1
Sales       | Account Executive   |      1
Marketing   | Marketing Manager   |      1
HR          | HR Manager          |      1
IT          | Software Engineer    |      1
Marketing   | Content Specialist   |      1
Finance     | Financial Analyst    |      1
(12 rows)
```

```
postgres=# select department, count(*) from employees group by department having count(*) > 2;
 department | count
-----+-----
Sales       |      3
IT          |      3
(2 rows)

postgres=# INSERT INTO employees (department, job_title)
postgres=# VALUES
postgres=#      ('IT', 'Software Engineer'),
postgres=#      ('IT', 'Software Engineer'),
postgres=#      ('IT', 'Software Engineer'),
postgres=#      ('IT', 'Data Analyst'),
postgres=#      ('IT', 'Data Analyst'),
postgres=#      ('IT', 'DevOps Engineer'),
postgres=#      ('HR', 'HR Manager'),
postgres=#      ('HR', 'Recruiter'),
postgres=#      ('HR', 'Recruiter'),
postgres=#      ('HR', 'Recruiter'),
postgres=#      ('Finance', 'Accountant'),
postgres=#      ('Finance', 'Accountant'),
postgres=#      ('Finance', 'Financial Analyst'),
postgres=#      ('Marketing', 'Marketing Manager'),
postgres=#      ('Marketing', 'Content Specialist'),
postgres=#      ('Marketing', 'Content Specialist'),
postgres=#      ('Sales', 'Sales Representative'),
postgres=#      ('Sales', 'Sales Representative'),
postgres=#      ('Sales', 'Sales Manager'),
postgres=#      ('Sales', 'Sales Manager'),
postgres=#      ('Sales', 'Account Executive');
INSERT 0 21
postgres=# select department, job_title, count(*) from employees group by department,job_title having count(*) > 2;
 department |      job_title      | count
-----+-----+-----
IT          | Data Analyst        |      3
HR          | Recruiter           |      4
Finance     | Accountant          |      3
Sales       | Sales Manager       |      3
Sales       | Sales Representative |      3
IT          | Software Engineer    |      4
Marketing   | Content Specialist   |      3
(7 rows)
```

Task 4

```
[postgres=# create table orders(orderID serial, customerID int, orderAmount int, orderDate date);
CREATE TABLE
postgres=# INSERT INTO orders (customerID, orderAmount, orderDate)
postgres=# VALUES
postgres=#      -- 2024
postgres=#      (101, 250, '2024-01-15'),
postgres=#      (102, 450, '2024-01-22'),
postgres=#      (103, 120, '2024-02-10'),
postgres=#      (101, 300, '2024-02-18'),
postgres=#      (104, 750, '2024-03-05'),
postgres=#      (105, 200, '2024-03-19'),
postgres=#      (102, 600, '2024-04-11'),
postgres=#      (103, 350, '2024-04-25'),
postgres=#      (106, 900, '2024-05-08'),
postgres=#      (101, 150, '2024-05-20'),
postgres=#      (107, 500, '2024-06-14'),
postgres=#      (104, 400, '2024-07-03'),
postgres=#      (105, 650, '2024-07-21'),
postgres=#      (108, 275, '2024-08-16'),
postgres=#      (102, 800, '2024-09-10'),
postgres=#      (106, 350, '2024-09-28'),
postgres=#      (109, 1000, '2024-10-12'),
postgres=#      (103, 450, '2024-11-05'),
postgres=#      (107, 700, '2024-11-19'),
postgres=#      (110, 325, '2024-12-15'),
```

```
[postgres=# select customerID, sum(orderAmount) from orders where extract(year from orderDate) = 2025 group by customerID having sum(orderAmount) > 1500;
 customerid | sum
-----+-----
          101 | 1600
          102 | 1650
          104 | 1550
          106 | 2150
          108 | 1700
(5 rows)
```

Task 5

```
postgres=# create table employees(employeeID serial, employeeName varchar(50), department varchar(50), salary numeric, joiningDate date);
CREATE TABLE
postgres=# INSERT INTO employees (employeeName, department, salary, joiningDate)
postgres=# VALUES
postgres-# ('John Smith', 'IT', 75000, '2019-02-15'),
postgres-# ('Sarah Johnson', 'IT', 82000, '2020-06-10'),
postgres-# ('Michael Brown', 'IT', 68000, '2021-01-25'),
postgres-# ('Emily Davis', 'IT', 91000, '2019-11-03'),
postgres-# ('David Wilson', 'IT', 78000, '2021-07-18'),
postgres-# ('Jessica Miller', 'HR', 55000, '2019-04-22'),
postgres-# ('Daniel Moore', 'HR', 62000, '2020-09-14'),
postgres-# ('Laura Taylor', 'HR', 58000, '2021-03-08'),
postgres-# ('James Anderson', 'Finance', 72000, '2019-01-12'),
postgres-# ('Linda Thomas', 'Finance', 85000, '2020-05-27'),
postgres-# ('Robert Jackson', 'Finance', 79000, '2021-02-19'),
postgres-# ('Patricia White', 'Finance', 93000, '2019-08-30'),
postgres-# ('Christopher Harris', 'Marketing', 65000, '2020-01-17'),
postgres-# ('Barbara Martin', 'Marketing', 71000, '2019-06-05'),
postgres-# ('Matthew Thompson', 'Marketing', 68000, '2021-05-21'),
postgres-# ('Anthony Garcia', 'Sales', 60000, '2019-03-11'),
postgres-# ('Jennifer Martinez', 'Sales', 74000, '2020-07-29'),
postgres-# ('Mark Robinson', 'Sales', 69000, '2021-04-16'),
postgres-# ('Susan Clark', 'Sales', 88000, '2020-10-12'),
postgres-# ('Steven Rodriguez', 'Operations', 67000, '2019-12-01');
INSERT 0 20
```

```
postgres=# select department, string_agg(employeeName, ', ' ) from employees where extract(year from joiningDate) > 2020 group by department having avg(salary) > 70000;
 department | string_agg
-----+-----
 Finance    | Robert Jackson
 IT         | Michael Brown, David Wilson
(2 rows)
```